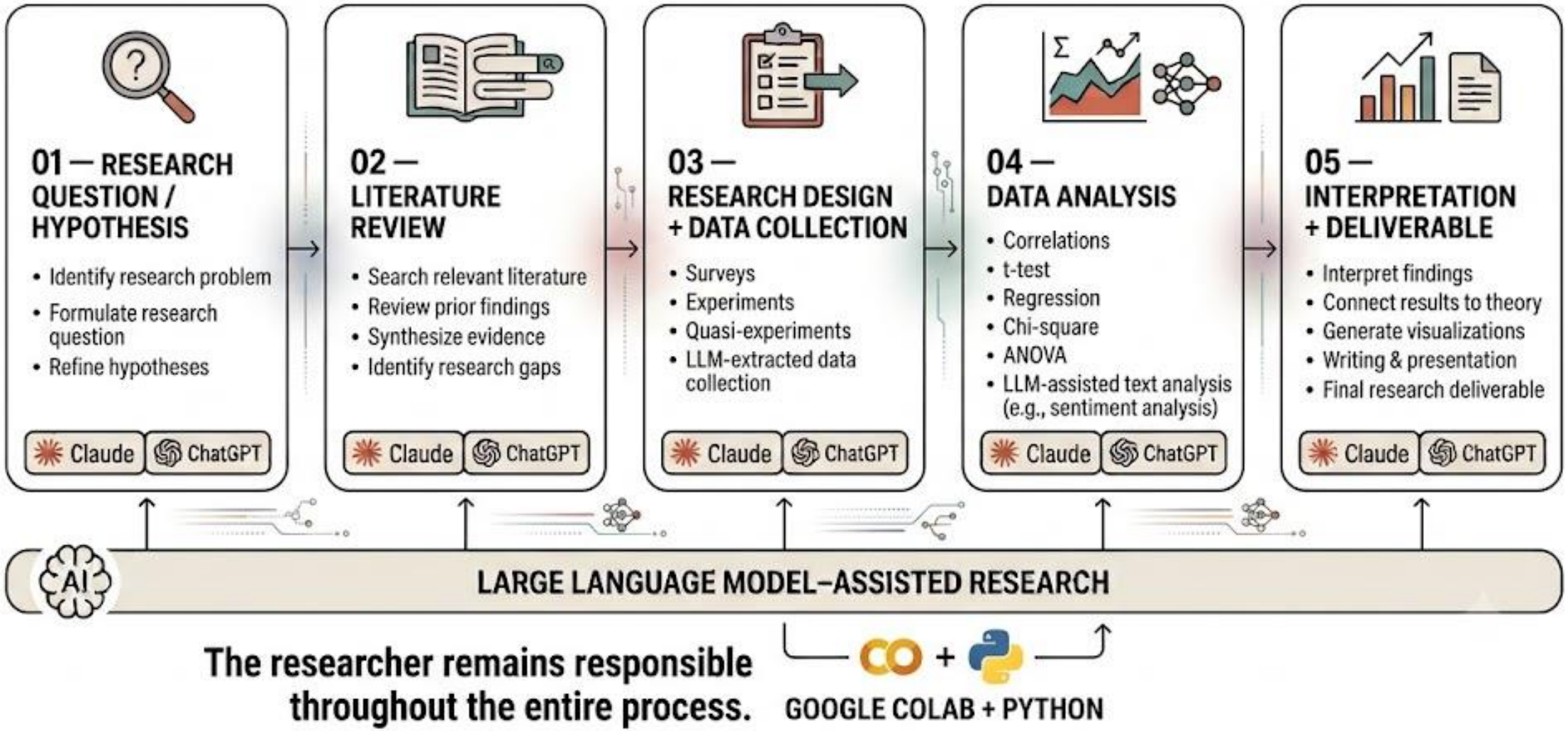
Computational Text Analysis

JOUR 701 • NOVEMBER 17, 2026

Research Methods in the Age of AI



The researcher remains responsible throughout the entire process.

GOOGLE COLAB + PYTHON

What is computational text analysis?

NEW CONCEPT

01 A family of methods for examining patterns in written or spoken language using computers.

02 Inputs: survey responses, comments, headlines, transcripts, or articles.

03 Outputs: counts, indicators, categories, scores, or patterns that researchers interpret.

Text → documented transformation → measure → interpretation.

Why text analysis in mass communication?

WHY IT MATTERS

01 Our field studies language: headlines, news stories, campaigns, comments, and audiences.

02 Surveys tell us how people rate a message. Open-ended responses can explain why.

03 Computational tools help analyze many texts using documented, repeatable rules.

Research question first. Tool second.

Remember Block 1: NLP, LLMs, and tokens

BLOCK 1 RECAP

01 NLP = methods for working with human language using computers.

02 A token is a unit of text; different tools divide text into units differently.

03 An LLM learns patterns from large amounts of text and can generate or classify language.

NLP is broader than LLMs. Not every text task requires AI.

‘From Rules to Large Language Models: The Evolution of NLP’

1950s–1980s: Rule-Based NLP

'Short Label: 'Humans write explicit language rules.'



Grammar and spell-checker



ELIZA-style chatbot



Machine translation rules



Telephone menu / IVR

1980s–2000s: Keyword-Based and Statistical NLP

'Computers learn patterns from word counts and probabilities.'



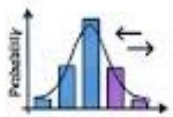
Spam filter



Search engine



Predictive text



Statistical machine translation



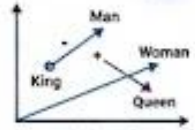
News classification

2000s–2016: Machine Learning and Neural NLP

'Artificial neural networks process language context.'



Recurrent Neural Networks (RNNs)



Word Embeddings (Word2Vec)



Sentiment Analysis



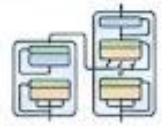
Named Entity Recognition (NER)



Question Answering systems

2016–2018: The Rise of Transformers

'The 'Transformer' architecture enables better sequence learning.'



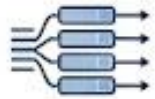
Transformer architecture



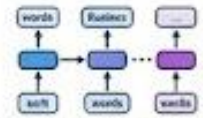
Self-Attention mechanism



Pre-training and fine-tuning



Parallel processing



Contextualized word embeddings (BERT)

2018–Present: Large Language Models (LLMs)

'Massive models pre-trained on vast internet datasets.'



Massive scale



Generative AI

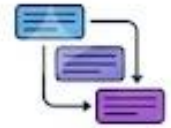




GPT-4 and advanced models



Code generation



Few-shot learning

Three approaches you already know

Simple rules

Find a word, count terms, or match a keyword.

Pretrained NLP model

Use an existing classifier to assign a label.

Generative LLM

Give a prompt and ask for a structured interpretation.

Today we start with transparent rules—not an LLM.

One text answer may provide more than a numeric rating

WORKED EXAMPLE

“I liked the information, but I felt uneasy about how personal it was.”

One response contains approval AND discomfort.

Return to the Block 3 experiment

SAME RESEARCH STORY

DESIGN

180 participants; randomized to Control or Personalized message.

EXISTING OUTCOMES

message_trust (numeric), clicked_article (Yes / No).

NEW QUESTION

“In your own words, how did you feel about the message?”

What columns are in our teaching dataset?

Variable	Type	Example
condition	Categorical	Control / Personalized
message_trust	Numeric	1–5-style rating
clicked_article	Categorical	Yes / No
open_response	Text	“It felt helpful but intrusive.”

Our guiding research question

WORKED EXAMPLE

How do people describe the messages they saw?

Start by describing the words—not by testing a hypothesis.

A simple workflow

1 • Define

What exactly are we trying to learn from the responses?

2 • Process

Inspect, clean, split, and count text consistently.

3 • Interpret

Return to the original wording and research context.

No computer can decide whether a measure answers your question for you.

First descriptive measure: response length

WORKED EXAMPLE

“The message was useful but felt too personal.”

Count how many words appear in each response.

Word count measures length—not positivity, understanding, or credibility.

Before counting: clean text carefully

01 Make upper- and lowercase consistent when counting the same words.

02 Split text into individual words (tokenization).

03 Remove common stopwords only when they do not matter for the question.

For sentiment, “not” can reverse meaning: do not remove it casually.

Word frequency: useful but limited

COMPARE

What it CAN say

Which terms occur most often?

Which words appear more in Control vs. Personalized responses?

What it CANNOT say

Whether the speaker feels positive.

Whether “trust” means “I trust” or “I do not trust.”

Most frequent ≠ most important.

A keyword rule is an operational definition

WORKED EXAMPLE

Does a response contain “trust” or “trustworthy”?

Store True or False for each response.

“I trust this post” and “I do not trust this post” both match.

Link the new text variables back to Block 3

SAME METHODS, NEW MEASURES

X	Y	Question	Test
condition: 2 groups	word_count: numeric	Mean length differs?	t-test*
condition: 2 groups	mentions_trust: Yes / No	Keyword mentions related?	chi-square*
word_count: numeric	message_trust: numeric	Do scores vary together?	correlation*

***After checking assumptions, data quality, and whether the measure is valid.**

Guided Colab: open and inspect

STUDENT ACTIVITY

12 MIN

1. Upload the provided synthetic CSV.
2. Identify the `open_response` column and read five examples.
3. Count words and compare average length by condition.

Guided Colab: frequencies and keywords

STUDENT ACTIVITY

14 MIN

1. Run the top-word frequency table and chart.
2. Compare the most common words by message condition.
3. Create and inspect the `mentions_trust` True/False indicator.

Which words or examples would you check by hand?

Do not confuse counting with understanding

COMPARE

“The message was not trustworthy.”

Rule: trust keyword = TRUE

But the meaning is negative.

“The message was trustworthy.”

Rule: trust keyword = TRUE

But the meaning is positive.

Same keyword. Opposite meaning.

Think–pair–share: apply the idea

STUDENT ACTIVITY

6 MIN

1. Name one variable we created from text.

2. Explain what that variable measures—and what it does NOT measure.

3. Choose a Block 3 test and justify it using the X and Y variable types.

Share one example of a misleading keyword count.

What comes next: sentiment classification

NEXT CLASS

01 A frequency table describes which words appear.

02 Sentiment classification assigns a category such as positive, neutral, or negative.

03 Next class we compare human coding, a lexicon rule, and a Hugging Face classifier.

The classifier’s label is a measurement to evaluate, not a fact.

Exit ticket

EXIT TICKET

1. What does computational text analysis help a mass communication researcher do?

2. Why can a frequent word or keyword match be misleading?

3. Which existing statistical test could use a text-derived measure?